

Element: An Ambient Display System for Evoking Self-Reflections and Supporting Social-Interactions in a Workspace

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ABSTRACT

In this paper, we present the design of *Element*, an ambient display system that displays the user's emotional engagement level through color and shape change on a light fixture. Given the increasing number of people who experience exhaustion or burnout in professional contexts, our goal is to design an interface that helps people keep track of their engagements through an ambient display of real-time physiological data. We conduct two qualitative studies to investigate its possible benefits and users' concerns from an individual perspective and social contexts with multiple users. The result shows that our design can not only aid users' self-reflections and discovery on the intrapersonal level, it also shows the potential to help participants understand others' emotional engagement and provide vital support when needed. Interestingly, our findings also highlighted the ethical concerns of displaying personal data in a public space. Reflecting on this crucial challenge, we also propose three social guidelines for helping users establish positive mindsets and cultures of using *Element* appropriately in sociotechnical environments.

CCS CONCEPTS

• **Human-centered computing** → *User interface design; Empirical studies in interaction design.*

KEYWORDS

Self-awareness; Job Burnout; Collocated Social Interaction; Ambient Display

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1 INTRODUCTION

Physical and emotional exhaustion has become a common occupational phenomena in many domains worldwide. Employees experiencing disengagement and job dissatisfaction often encounter the familiar social and chronic problem known as *job burnout*. In 2019, a study with 7500 full-time employees showed that 23% of participants reported feeling burned out at work very often [21]. Another 2021 study shows 38% expressed they were "burned out to a great extent." [10]. Job burnout is a psychological syndrome that involves a prolonged response to chronic emotional and interpersonal stressors in the workplace [12]. The causes are multidimensional, and practitioners often use Maslach Burnout Inventory [13] to assess an individual's experience of burnout through observing more than 20 items.

For ordinary people, it is often hard to understand and keep track of one's engagement and workloads in daily practice. Without knowing when the possible symptom (e.g., disengagement) starts to manifest, it is not easy to tailor solutions in a timely and relevant manner. Our research aims to explore designs that could enable users to become conscientious of their own emotional engagement at work and react appropriately before the situation becomes harmful mentally or physically. Several HCI and CSCW studies (such as [20]) have already shown that social and cultural factors are essential in office contexts, and social bonding can be an efficient intervention for changing ones' behaviors. Therefore, we want to understand how people think about displaying individual data in a working space, their benefits, and how ethical norms and privacy should be considered in designing sociotechnical systems.

This paper presents our first design outcome: *Element*, a light fixture that displays the user's emotional engagement level in the workplace through color (and shape) change. In the HCI community, Galvanic Skin Response (GSR) technology has been successfully utilized in measuring users' cognitive load levels [7, 24] and emotional arousal [16]. Our research applied commercial GSR sensors to monitor a user's emotional arousal and analyze the data to infer their emotional engagement levels as an indicator of their mental state in near real-time. We are aware of the ethical and privacy issues of collecting and displaying personal data in a shared space. Correspondingly, we purposely chose analogical and ambiguous visualizations to foster self-reflection rather than representing the data with numbers and graphs. In addition, we also followed the research-through-design method [31] in the way of not finding a specific solution, but for exploring the broader considerations of designing a sociotechnical system involving multiple people. Our first idea, *Element 1*, is a table lamp that represents a user's engagement



Figure 1: The vision of our final design of *Element 2* in a workspace. Informed by a user's biodata and the inference of their emotional engagement, *Element 2* represents the result with color and shape-changing for evoking users' self-reflections. It can also foster social support by inviting their interpretations of the open-ended expressions. A video demonstration is available on <https://vimeo.com/251813730>.

level with color change. By reflecting on its design with the users' feedback and reflections, we pivoted the table form to a pendant lamp (see Fig.1) that could better encourage social interactions [29] among colleagues in a workspace.

Our contributions are threefold. Firstly, we expand the design space of an ambient display with two designs of light fixtures and integrated GSR sensors. Rather than informing users precisely, we use ambiguity design to foster self-reflection and social interactions. Secondly, we report the empirical findings with the two qualitative studies with *Element 1* to investigate users' experiences and possible concerns. The results show that our design has the potential to help users establish a healthier working atmosphere on two levels. On an intrapersonal level, it aids self-discovery and awareness of energy depletion or exhaustion. In an interpersonal context, our design could help people understand others better and thereby encourage social interaction whenever needed and appropriate. Thirdly, we demonstrate how an information system should be designed with a broader scope, encompassing ethical norms and privacy concerns. Specifically, we proposed three social guidelines for people to establish the right mindsets in using the *Element* system properly. We believe that a collective understanding of ourselves and other colleagues can improve day-to-day interactions and increase the sense of community. More importantly, it can provide a foundation for building healthier well-being and preventing job burnout in the future.

2 RELATED WORKS

2.1 The Emerging Technologies for Sensing a User's Physiological Status

Many emerging technologies have been developed to measure a user's cognitive load levels, such as eye-tracking, facial expression analysis, electroencephalography, Galvanic Skin Response

(GSR) sensors [9], etc. Among them, GSR is a measure of the ever-changing conductance of our skin. The electrical properties of our skin change when our sweat glands respond to various factors, such as changes in psychological or physiological arousal [2]. Previous studies have shown the value of skin conductance in measuring users' cognitive load level in using information systems [24], doing creation tasks [7], supporting individual reflections on everyday emotional experience [26], etc. Empowered with advanced algorithms, the data could also be used to generate meaningful inferences about a user's emotional status [11]. Therefore, we chose the GSR sensors to measure a user's emotional engagement in our research.

2.2 Ambient Display Systems

Ambient Display is an emergent research field that has gained increasing interest since the early 2010s. In a study on the subject, Moore [15] observed that ambient information systems had recently begun evolving towards an application he coined as 'persuasive visualization', and that these displays, "could increase human awareness by conveying [...] relevant issues with a relevant and timely environmental context." Two recent studies stand out in the research related to this topic. In the first *FlowLight* [32], the keyboard and mouse activity and the status of specific applications on the users' computer were tracked in an IT office to inform a personal ambient display. By switching a LED light from green (available) to red (busy), *FlowLight* informed colleagues when its user was 'in the flow' and should not be disturbed. This automatic process allowed for transitions into engaged work that were not punctuated by manually putting up a 'do not disturb' sign. Reduced in-person interruptions represent not only a change in workplace culture, but account for a measurable increase in performance and profit.

The second work examines some of the personal and social implications of displaying personal data publicly. *MoodLight* [25] utilized data collected with electrodermal activity sensors (similar to GSR sensors) to support stress management with university students who felt it challenging to practice relaxation techniques. The researchers concluded that the usage of ambient light provided a promising medium to cultivate healthy self-awareness in social contexts. It gave a user momentary feedback with an emphasis on the present moment particularly suited to supporting mindfulness practice. However, it was also reported that the representation of the internal emotional state was not a welcome discovery for some students, especially for the negative feedback loop that occurred when they experienced high levels of arousal that triggered the light color to red. Responding to those prior findings, our research aim to design evocative visualizations with balanced openness and ambiguity for inviting rich interpretations [23, 27].

2.3 Social Interactions and Privacy Concerns

Apart from empowering the individual to have better awareness of their statuses, design interventions could also support an interpersonal approach to leverage social bonding and interactions [17, 20]. Whether personal or group-based, tracking and reflecting on affective states has been shown to bring enjoyable experiences and help users become healthier [3, 14, 25, 27]. For instance, [6]

favors moving out from pure modalities of e-coaching (technology) to a more hybrid approach involving human social expertise in real-time. Their social perspective inspires us to consider the public acknowledgment of data and use data visualization to not only help the primary user, but also facilitate users to support each other. While we are excited to incorporate social interactions in our design exploration, we are aware of the tensions of interpersonal privacy in sociotechnical environments [18]. Instead of informing people about something unusual, we aim to promote collaborative sensemaking [19, 30] and collocated social interactions [17]. Our research took a broader scope in accordance with ethical norms and privacy concerns. Specifically, we purposely chose analogical and ambiguous visualizations to foster open-ended interpretations [8]. Further, the physiological data being represented is itself ambiguous; emotional arousal measured through GSR is simply capturing the intensity of the users' emotions, but is not able to attribute positive or negative valence to them. This distinction points to an important interpersonal dynamic which we come back to in presenting our three social guidelines for *Element*. We used the research-through-design method [31] to investigate the possible social effects with the ambient display of this data, and to understand how a sociotechnical system should be designed correspondingly.

3 DESIGN PROCESS

We took a research-through-design [31] approach in researching the broader design challenges (e.g., data privacy and ethical considerations) through making experiential prototypes and using them to investigate people's thoughts. The designing and evaluation activities were intertwined within the entire design process. In this paper, we primarily present the first iteration and insights learned from the two qualitative studies with ten participants in total. The user study and main findings are described in Section 4.

3.1 The Design of *Element 1*

In the beginning, we explored multiple ideas with the mediation theory [28] by investigating how technology could play a mediating role in users' working contexts. We also research the technical feasibility through quick prototyping, system implementation, and accuracy validation. As a result, we developed a table light fixture, called *Element 1*, to provide people awareness of their own emotional arousal with the real-time biodata gathered from GSR sensors embedded in a wristband. In a social context, we envision that a light fixture could also subtly attract users' attention to invite and encourage collocated social interactions among people in close proximity [17]. In the long term, it could strengthen emotional bonds and increase the sense of community.

This *Element 1* system consists of three components (see Fig. 2). First, a GSR sensor was connected to Arduino to measure the subtle signals from a user's skin conductance. We follow [26]'s approach in collecting and processing the data with an Arduino Uno. Instead of displaying the sensor values directly, we mapped the calibrated data to RGB value for setting light color on *Element 1*. Our intention of choosing an ambiguous design is to evoke users' self-reflection and invite rich interpretations [5, 8, 23]. To eliminate the human tendency of direct reactions on specific colors used in MoodLight[25], we chose the warm (yellow) to cool (blue) to

represent the high to low arousal status. The midpoint between these two extremes was a desaturated white light. Animations were also implemented for smooth transitions in step steps that engaged whenever values moved between certain ranges. An image of our prototype is shown in Fig. 3).

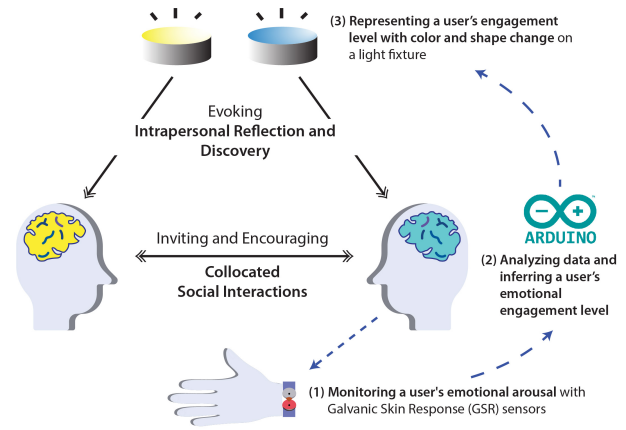


Figure 2: The *Element* system consists of three components, including (1) the GSR sensors attached to the user's wrist, (2) the computer algorithm is running on an Arduino board for analyzing the biodata and transferring to a numeric value, and (3) the cylindrical light fixture that could display the degree of arousal status with the color from the cool-warm color spectrum.

3.2 Functionality Validation

Before using our prototype to probe users' experiences, we validated the quality of the data gathered from our system by comparing it to data collected with EdaMove 3, a commercial sensor produced by Movisens (www.movisens.com). One of the authors attached the sensors of both devices simultaneously, one on each wrist. A five-minute workout routine was then carried out continuously in two hours, with periods of rest and deep breathing between exercises to try and make contrasting peaks and valleys in the data. The raw data from both devices were imported to Excel for comparison. After calibrating the two data sets, it was found that the relative

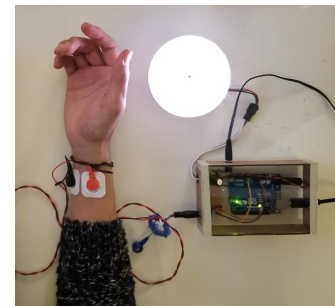


Figure 3: The working prototype of *Element 1*.

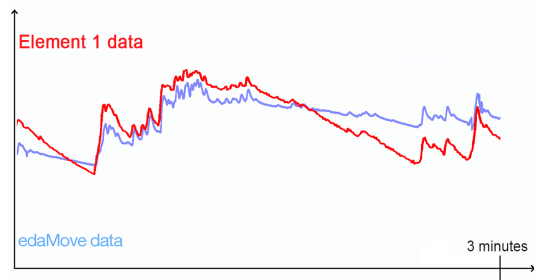


Figure 4: The visual comparison of the data collected by our *Element 1* prototype and the commercial product, EdaMove 3. The results were highly correlated ($r = 0.92$). Our system can track and analyze the changes in users' arousal status and inform them appropriately.

peaks and valleys were highly correlated ($r = 0.92$, the comparison of the visual chart is displayed in Fig. 4). Although the signal of *Element 1* was less stable, through analyzing the peak moment with the activity log made by the tester, we have a certain confidence that our system is capable of capturing the overall tendency of change in a user's arousal states.

4 THE USER STUDY WITH *ELEMENT 1* AND MAIN FINDINGS

In order to investigate the efficacy of our design in fulfilling the two different purposes of evoking self-reflection and collaborative sensemaking aforementioned, we conducted two studies to investigate the user experiences. In the first diary study, we recruited a Ph.D. candidate (male aged 25) from our department to use *Element 1* for one week. Before the test started, we calibrated our sensor with the commercial EdaMove 3 and instructed him how to wear the sensors properly. The participant was then left to use *Element 1* as many hours per day as he wished, with an apparent preference for using the prototype as much as possible. We designed and gave a small journal booklet and encouraged him to write down his activities and the reflections evoked from *Element 1*. To engage the participant in logging his user experiences, we designed several small activities with visual elements, including registering his momentary feelings on the user experience curve, quickly writing down their reflections with the sentence completion methods, and reviewing his daily with Day Reconstruction exercise. After the study was completed, a semi-structured interview was conducted to gather the participant's insights as a user and design researcher.

In the second group study, we recruited nine students from our department (five males and four females, aged 22-25) to examine the social interactions and collaboration perspectives. We set up a group study with three participants per time to probe their experiences and reflections. Before the testing started, we attached the wristband to each participant and calibrated the settings per person. A prototype was placed in front of each user. At the same time, they all could also see the others' prototypes in their periphery. In order to trigger the changes in their arousal statuses, we refer to Liu et

al.'s [11] approach of using video materials to elicit users' emotions and purposely chose three video clips from films. Each clip was meant to elicit a range of emotional responses, and the length was between four and eight minutes each. After each clip ended, there was a break of five minutes. Participants were asked to check the lights shown on each prototype and discuss their thoughts together with the researcher. When all the sessions were completed, we conducted a semi-structured interview to understand their experiences, probe their reflections, and discuss improvement ideas and future applications.

4.1 The Findings of the One-Week Diary Study

At the beginning of the study, the participant asked himself, "what does the light color mean?" He first admitted that he thought of yellow as only stress, only later realizing that it also reflected a more positive state of concentration. Understanding this, he then started seeing *Element 1* more to reflect what he may be feeling. On one occasion, for instance, he saw the light turning yellow and paused to assess, stating that he realized, "I'm not stressed but not particularly focused ... Oh, I'm getting hungry. Actually, it's that". There were other instances of self-discovery. During a late night of stressful work, he took a break by looking at his phone for pictures and talking to his girlfriend. When he did so, he was surprised to find that *Element 1* drastically changed to blue, mirroring his relaxation. The change was so abrupt that he even doubted that the sensor was working until he resumed his work, and the reading came back up to a slight yellow. He later confirmed that his de-stressing break (and the sensor) had worked when he recreated the experience on another day. He stated that he now trusts more in these breaks because he saw their effect visualized in *Element 1*.

Although he did not gather meaningful comments from coworkers to his engagement display, he did extrapolate to what it may be like to have an *Element 1* in a more social setting. "It makes me more vulnerable, but I can see it also as a way that people around me are prone to ask about how I'm doing," he stated. He seemed optimistic about it, also remarking that he would see it as a positive challenge to include his supervisor in his stress in a supportive capacity. He said: "if [my supervisor] could see that I was stressed about something anyways, from my face, but it was not crystal clear [...] he could see 'oh, this is yellow, what's going on [...] is it something at work?' At the same time, he also stated that it would be good to have a way to turn it into a neutral setting because "there are some situations where I need to bluff."

Overall, our prototype worked well for the entire week. The participant interpreted the color associations are similar to the designer's intentions. He saw blue as sharp and bright while the yellow was 'tired.' He also associated the warmer yellow to how applications (such as f.lux [1]) gradually make computer and phone screens warmer in color temperature as the sunsets. Since this is done to protect the circadian rhythm from disruptive blue light in the evening, it is associated with ending the day and sleeping well. When working late, the participant reported turning off *Element 1* to keep himself focused and in 'work mode.' The yellow light for a high engagement at such times may then be disruptive, as it is opposite to that 'work mode' setting and reminiscent of nighttime and rest.

4.2 The Findings of the Group Evaluation

First, the visualizations of their own engagement status were understood well in almost all instances; little clarification was needed. Their interpretations of the colors were well aligned with their momentary feelings and the designers' intentions, as we found in the diary study aforementioned. This shows our design has good quality in ambiguity because it evoked a certain range of responses rather than random [8]. When the conversation shifted to potential benefits, many participants became excited and optimistic, a status that was reflected by *Element 1* and was itself remarked upon several times. Regarding self-discovery and reflection, all participants could pinpoint concrete benefits and agree with each other's ideas, including better adjusting work/break habits, knowing oneself, maintaining a better work-life balance, etc. While inquiring about their present work habits, all but two participants acknowledged that they have difficulty in staying aware of their efficiency and output. Some reported that they often forgot to do momentary reflections. When they tried to reflect, it usually was weeks later or when the physical discomfort had already accumulated into something more serious. "I feel my shoulders start to hurt," said P4. This delay in identifying the situation and problem is one of the main things that could be helped by increasing self-awareness through *Element 1*. The real-time feedback could trigger people's attention and involvement in reflecting on their behaviors. One participant identified more profound self-discovery benefits potentially available. Looking to narrow down her upcoming career choices, P4 remarked that an objective measure would be helpful in revealing which types of work most engage her.

While we asked participants to reflect on their chatting or implicit interactions during the breaks between the video clips, they related their experiences to the collaborative benefits in their daily workings. They agreed with the system's advantages of getting to know other colleagues' emotional engagements and possible feelings of stress. They came out with several concrete benefits of having *Element 1*, such as not interrupting colleagues, supporting coworkers, and building sustainable working culture. Relating to the self-discovery, three participants also shared the common cases that they were too engaged in their tasks to pay any attention to physical or mental fatigue signals. In those situations, they envisioned that *Element 1* could enable collaborative sensemaking by attracting other colleagues' attention and providing meaningful data to discuss how they can resolve the issues together. If we extended this application to a higher level in team management, two participants provided possible ideas based on their manager experiences. For instance, P2 was managing a team with ten people in a small enterprise at that time. She remarked that "for a manager, it would be helpful to know which types of tasks are more compatible with which employees so that work can be assigned more appropriately." The experience with *Element 1* gave her an idea of collecting their engagement data with different assignments and using that to find their optimal matches. As a result, this could improve the collective sense of efficacy as a team and company. Although those ideas are all very inspiring and exciting, a concern regarding privacy in disclosing personal data in public spaces manifested in discussions within every group session. Almost all participants expressed concern about revealing their interpersonal data in the public space.

Surprisingly, two users also expressed a strong interest in knowing other people's emotional engagements. Acknowledging this duality, P5 remarked, "if [a colleague] is anxious or something [...] then maybe I could help! In that perspective, maybe it's nice to show my own engagement as well." In the next session, we will extend this topic and discuss strategies from literature which guided us in improving our design concept.

5 THE DESIGN OF SOCIAL GUIDELINES AND ELEMENT 2

Through the two qualitative studies conducted with *Element 1*, we saw the potential of our design concept. *Element 1* shows promising possibilities for integrating the bodily and cognitive interactions for triggering individual reflections [8]. This could improve personal ability in intensifying self-regulation for alleviating job burnout promptly [29]. On the other hand, we also learned that most participants had concerns about disclosing personal biodata in a public space. In order to tackle this ethical and privacy challenge, we see the advantages of integrating social/cultural interactions in our design considerations [8]. Inspired by Hook et al.'s [8] view on privacy as a negotiation between parties, we broaden our design concept from an artifact to incorporate setting up proper social cultures among the group of people. We advise the following three guidelines to help people establish appropriate mindsets and negotiate their wishes of using the system collaboratively:

- (1) Everyone is different: People's bodies react differently to stimuli, and people work most effectively at different stress / eustress levels.
- (2) High or low doesn't mean good or bad: The reading is highly ambiguous. Users should discuss the signals together rather than making judgments directly.
- (3) Context matters: To correctly interpret the display, one must also be familiar with the situation, the person, and other contextual cues.

We envision that those guidelines could foster the dialectic process [18] for building a supportive culture in the sociotechnical environment.

To better invite and encourage social interactions among colleagues, we also redesigned the light texture and user interfaces. The latest design of *Element 2* and the link to a video demonstration can be found in Fig.5. We elaborate on the three primary considerations of our design rationale as follows.

Firstly, we realized that the table lamp modality of *Element 1* was sometimes distracting when it was put on a user's working desk. Considering the space constrain a user has in a typical office layout, we see the opportunity of moving it into the user's periphery, such as a pendant or wall light. Through light's glanceability [4, 22], we believe that it could still evoke users' awareness and reflections when they perceive the color changes through peripheral perceptions.

Secondly, we envision the pendant form can provide two additional benefits. Designers can integrate the ambient display into the task light available in most workplaces, and the information can be perceived by colleagues in physical proximity and thus encourage social interactions. Increasing the publicity of personal data rather than constraining its visibility is challenging. Noting [8], we would

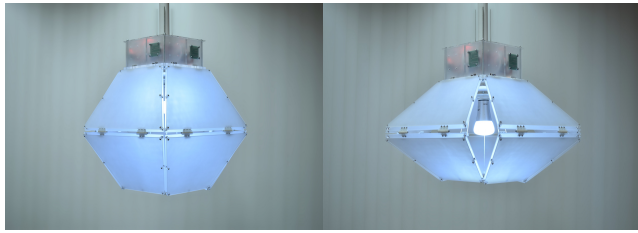


Figure 5: The demonstration of shape change in the prototype of *Element 2*. A video can be accessed with the supplementary material or via <https://vimeo.com/251813730>.

intend for this intervention to be installed in a supportive culture where proper privacy negotiations have preceded.

Thirdly, we added a shape-changing modality to form a multi-dimensional space for open-ended interpretations. The primary user is able to customize the representations of the light fixture by setting up the mapping of sensing data and light parameters on a mobile phone application. To help groups of people learn how *Element* can benefit their well-being and community, open communication should be encouraged and maintained. Such communication should include negotiations on when and what to share and how to use the system collectively. Together with the three social guidelines proposed above, we believe users can establish a positive culture of proactive communal support..

6 CONCLUSION

In this paper, we describe the design of *Element 1* and *2* which utilized the real-time biosensor data to present a user's emotional engagement with open-ended representations of color and shape on light fixtures. Based on the results of two qualitative studies, we found that our design can aid working culture on two levels: on an intrapersonal level, *Element* could facilitate self-discovery on his own working habits and engagement; in the interpersonal context, our design has the potential to help people understand one another better and thereby can provide social supports proactively when someone shows physical or emotional exhaustion. Compared to the findings of previous studies, such as FlowLight [32] and MoodLight [25], our design of *Element 1* comprehends the benefits of using ambient displays as an effective interaction in triggering self-awareness and inviting collocated social interactions [17]. Unsurprisingly, we also see that our project moved further and touched the line between the seemingly unique potential of biodata and the concerns of utilizing the data in sociotechnical environments. Inspired by literature and the participants' feedback, we extended our strategy beyond the artifact level and proposed three guidelines for helping people establish correct mindsets and positive cultures in using the *Element* system. Only when users build proper social/cultural interactions can our technical system function as well as we envisioned. Although it is not easy, we believe that an improved understanding of ourselves and our colleagues at work could promote people's wellbeing and aid in healthier engagements for preventing chronic burnout in the digital era.

7 LIMITATION

We realized that there were some limitations to the study. The number of participants in both studies could be increased. However, we believed that we already did a decent job gathering valuable insights for evaluating our designs. It also triggered us to look beyond the artifact level and inspired us to set up three social guidelines (see Section 5) on helping people establish a positive culture around using the system correctly. Lastly, due to the current restrictions of social distancing, we could not conduct the user experience evaluation in the actual context. We have set up a plan of investigating user experience with the *Element 2* design and guidelines aforementioned with six groups of people in our next step.

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